

CSE108 – Computer Programming Laboratory

Lab #6

Date: April 03, 2026

Part 0. Structural Requirements

To ensure code quality and prevent logic errors, your program must adhere to the following rules:

- **Distinct Functions:** Each task must be implemented as a separate function. Your `main()` must be menu-driven and call these functions.
 - **Separation of Concerns:** Functions must receive their inputs through parameters and return their results. Do **not** use `printf` or `scanf` inside any function other than `main()`.
 - **Library Constraints:** Use only `stdio.h`, `string.h`, and `math.h`.
 - **Array Traversal:** Use for loops to process array elements in sequence.
 - **Write your own main():** You are expected to write the `main()` function yourself, using a menu to call each part.
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Part 1 [50 pts] The “Word” Analyzer

Scenario: You will read 4 words from the user and find the longest one.

Data Input

In `main()`, use `scanf("%s", ...)` to read each of the 4 words from the user into a 2D character array declared as:

```
1 char words[4][20];
```

Function Prototype

```
1 void find_longest_word(char words[4][20], char longest[]);
```

Logic

- Use `strlen()` to determine the length of each string.
- Iterate through all 4 words to identify the longest one.
- Copy the longest word into the `longest` output parameter using `strcpy()`.

Output

In `main()`, after calling `find_longest_word()`, print the longest word and its length.

Example

```
Enter word 1:  apple
Enter word 2:  banana
Enter word 3:  fig
Enter word 4:  strawberry

Longest word:  strawberry
Length:  10
```

Part 2 [50 pts] Simple Word Search

Scenario: Using the same 4 words read in Part 1, check whether a specific word exists in the list.

Function Prototype

```
1 int search_word(char list[4][20], char target[]);
```

Logic

- Use a linear search to check each word one by one.
- Use `strcmp()` to compare `target` with each word in the list.
- Return the index (0, 1, 2, or 3) if found, or -1 if the target is not in the list.

Output

In `main()`, prompt the user to enter a search word, call `search_word()`, then print either "Word found at index X" or "Word not found".

Example

```
Enter a word to search:  banana

Word found at index 1
```

```
Enter a word to search:  mango

Word not found
```